2012 Integration Bee Solutions

int1. $2\log(\sqrt{x}-1)+2\sqrt{x}$

int2. $(4x^{5/4}\log(x))/5-16x^{5/4}/25$

int3. $2(\sqrt{x}-1)/\sqrt{1-x}$

int4. $-(9\sqrt[4]{x}+1)/\Big(18(\sqrt[4]{x}+1)^9\Big)$

int5. $\pi/4$

int6. $\sin^{-1}\Big((x+2)/\sqrt{5}\Big)$

int7. $1/12$

int8. 1

int9. 1005

int10. $2\tan^{-1}\sqrt{x+\frac{1}{x}+1}$

int11. $33/2-24\log(2)$

int12. $\sin(x)\log(x)$

int13. $\log\sqrt{1-1/x^2}$

int14. $1/2-\pi\sqrt{3}/12$

int15. $1/10100$

int16. $4/3$

int17. $6x^{1/6}-6\tan^{-1}(x^{1/6})$

int18. $(\log(x+\sqrt{x^2-1/2}))/\sqrt{2}$

int19. $2\tan^{-1}\sqrt{e^x-1}$ or $-2\sin^{-1}e^{-x/2}$

int20. $\frac{1}{4}\tan^{-1}(x^2/2)$

int21. $\Big(\cos(x)+\sin(x)\Big)/\Big(\cos(x)-\sin(x)\Big)$ or $\tan(x+\pi/4)$

int22. $\log \tanh(x/2) - x/\sinh(x)$

int23. $1192/45$

int24. $\log 8 = 3 \log 2$

int25. $2\sqrt{\sin(x)}$